



Threads

1. Create a Java program that finds prime numbers up to a specified number N using multithreading.

Requirements:

- Implement two threads:
 - NumberGenerator: Generates numbers from 1 to N and passes each number to the PrimeChecker.
 - PrimeChecker: Checks if each received number is prime and prints it if true.
- Implement a simple mechanism for thread communication (e.g., using a shared variable) to pass numbers between threads.
- Implement a method to determine if a number is prime.
- Ensure the PrimeChecker can terminate gracefully after all numbers have been processed.

Dr. Layla Abou-Hadeed

Eng. Ahmed ElSayed

Eng. Ali Hassan

Eng. Ahmed Ashraf

Eng. Seif Eldin Mahmoud

Eng. Miar Mamdouh

Eng. Ahmed AboEleid

Eng. Ahmed Essam

Eng. Menna Tullah Ihab

Eng. Abdelaziz Mohamed

Eng. Mahmoud Ramzy

Eng. Abdelrahman ElSayed



2. Develop a Java program that simulates a simple bank account system using multithreading.

Requirements:

- Create a **BankAccount** class with synchronized methods for:
 - **Deposit:** Adds money to the account.
 - **Withdraw:** Subtracts money from the account, with checks for sufficient balance.
- Implement two threads:
 - **DepositThread:** Deposits a random amount (between 10 and 100) every second for a total of 10 seconds.
 - **WithdrawThread:** Attempts to withdraw a random amount (between 10 and 100) every second for a total of 10 seconds, handling insufficient funds gracefully.
- After both threads have completed, print the final account balance.

Dr. Layla Abou-Hadeed

Eng. Ahmed ElSayed

Eng. Ali Hassan

Eng. Ahmed Ashraf

Eng. Seif Eldin Mahmoud

Eng. Miar Mamdouh

Eng. Ahmed AboEleid

Eng. Ahmed Essam

Eng. Menna Tullah Ihab

Eng. Abdelaziz Mohamed

Eng. Mahmoud Ramzy

Eng. Abdelrahman ElSayed



Required:

- You are required to solve the above question and deliver a zipped file containing the .java files.
- The deadline for delivery is **Friday, November 15, 2024, at 11:00 PM.**

Policy:

- You must work individually.
- Your code should be clean, readable, and commented on.
- Delivering a copy will be severely penalized for both parties, so delivering nothing is so much better than delivering a copy.
- No late submission is allowed.

Dr. Layla Abou-Hadeed

Eng. Ahmed ElSayed

Eng. Ali Hassan

Eng. Ahmed Ashraf

Eng. Seif Eldin Mahmoud

Eng. Miar Mamdouh

Eng. Ahmed AboEleid

Eng. Ahmed Essam

Eng. Menna Tullah Ihab

Eng. Abdelaziz Mohamed

Eng. Mahmoud Ramzy

Eng. Abdelrahman ElSayed